

ANCHOR MODE

What a vessel burns doing nothing

AMBESS — Anchor Mode Battery Energy Storage System

SAKSHAM-URJA · PROJECT SAMUDRA — MARINE ENERGY PLATFORM

THE HOURS NOBODY COSTS

Every operator knows what his vessel burns under way. Almost nobody knows what it burns at anchor.

Ships spend up to nine per cent of their time waiting at anchorage — and the auxiliary engines and boilers running through that time consume ten to fifteen per cent of the vessel's fuel. Nothing moves. Nothing is caught. No cargo is carried. The fuel burns anyway.

It is not a small part of the year. A research vessel's endurance is commonly planned on roughly half its sea time on station. A blue-water fishing boat holds over the grounds and lies at anchor between hauls for days at a stretch. A merchant ship waits its turn for a berth.

And it is the most expensive fuel on the ship. At anchor the generator carries a fraction of its rating, and a diesel running lightly loaded burns far more per unit of energy than one running near its best point. **You pay the highest price per kilowatt-hour at precisely the moment you need the fewest.**

None of it appears as a line item. It is buried inside the voyage fuel figure, where it has never been separated, never been questioned, and never been reduced.

The question

If the main engine can be stopped at anchor, why can't the generator?

The answer has always been that something has to keep the lights, the refrigeration and the electronics alive. That answer is now out of date.

Anchor mode

We named the product after the problem.

AMBESS — the Anchor Mode Battery Energy Storage System. It carries the vessel's hotel and auxiliary load at anchor and on station. **It does not drive the vessel, and that is the design intent rather than a limitation.**

A battery carries the load while the generator is **off**. The generator no longer idles for twenty-four hours producing a trickle of power. It runs a short block at high load — where it is efficient — puts energy into store, and stops. Where a solar array is fitted, part of that energy never comes from fuel at all.

The same work gets done. The refrigeration holds. The lights stay on. **The system maintains the designated hotel and auxiliary loads while the generator is stopped.** The difference is the number of hours the machine is running, and the load it is running at when it does.

This is not a theory and it is not a simulation.

The proof: 169 days at permanent anchor

We operate a 36.5 m platform that is, in effect, a vessel permanently at anchor. Its entire load is hotel and auxiliary. It is exactly the case described above, and it has been in continuous service since **15 November 2025**.

A written charge and discharge register kept by the crew covers **169 consecutive days**, 24 December 2025 to 10 June 2026, in **480 logged sessions**:

887.7	total generator hours across the period
5.25	generator hours per day
12–14	generator hours per day required before the system was fitted
24	hours of electrical availability, where there had been 12 to 14
zero	critical power failures logged

Conditions throughout: 38–45 °C, relative humidity above 85 per cent.

Nineteen handwritten pages. Not instrumentation we installed to prove a point — the operator's own record, which anyone is welcome to read.

What that means in fuel. At twelve litres an hour — the operator's own consumption figure for that set, applied to both sides — and diesel at Rs 99.82 a litre:

Against	Generator hrs/day	Litres/year avoided	Rs lakh/year
Standing orders at 12 hours	12	29,553	29.5
Standing orders at 14 hours	14	38,313	38.2
Round-the-clock power from the set alone	24	82,113	82.0

The last row answers a different question — *what would twenty-four-hour power from the generator alone have cost*. We show it beside the others so nobody has to work it out and wonder why we left it off.

And there is something the fuel figure does not capture. Standing orders called for the set to run twelve to fourteen hours a day, which meant the platform had no power for the other ten to twelve. It now has power for twenty-four. The fuel bill fell and the capability went up.

Where this is worth the most

Blue-water fishing

A 22–23 m Indian deep-sea trawler burns in the order of 25,000 litres of diesel per voyage. The auxiliary carries refrigeration continuously — a hold of 80 to 150 cubic metres needs 5 to 10 kW without pause — plus deck and fishing lights, hydraulics and electronics. That load runs whether the boat is trawling, drifting, hauling or lying at anchor. It is the part of the fuel bill that never stops, and the part rarely costed separately.

Research and survey vessels

Commonly planned on roughly half their sea time on station. Through those periods the main engines are idle or lightly loaded while auxiliaries carry laboratories, winches, instruments and accommodation. It is also the period when generator vibration and exhaust are least welcome — over the side of a vessel taking acoustic or water-column measurements.

Merchant vessels waiting for a berth

Anchorage congestion is a commercial cost measured in days. Nine per cent of the year, on a fuel share of ten to fifteen per cent.

Offshore support, dredgers, harbour craft, accommodation barges

Anything that spends its working life stationary with a live electrical load.

Why the principle holds — and it is not our claim to make

The International Maritime Organization has set this out in its own guidance on shipboard energy management.

On efficiency. Specific fuel consumption is at a minimum — efficiency at a maximum — typically between 70 and 90 per cent of an engine's maximum continuous rating. Below that, IMO states that under low load conditions the specific fuel consumption of the engine will increase and engine efficiency will reduce.

On what actually happens on board. IMO describes generators being run for long periods at less than 50 per cent load factor, naming standby periods among the conditions where this is sustained, and calls the result unnecessary simultaneous usage at low load factors and beyond requirements.

On the second cost, which is not fuel. IMO records that running diesel engines at low loads causes poor piston ring sealing, sub-optimum turbocharger performance, elevated thermal stresses and increased specific lubricating oil consumption — and concludes that it leads to more maintenance as well as higher fuel consumption.

We are not borrowing IMO's savings estimates. IMO's own measure is about running fewer generators at better load — a different and more modest intervention than stopping the generator altogether. What IMO establishes is the principle: **a lightly loaded diesel is an expensive and damaging way to make electricity, and reducing the hours it runs saves both fuel and maintenance.** Our register is what that principle looks like on one platform over 169 days.

Electrical integration and scope of supply

AMBESS is intended primarily for **selected hotel and non-propulsion auxiliary loads**. The allocation of loads, including any essential services, is determined during the vessel-specific electrical survey and class review.

It is designed to avoid interference with propulsion, steering and other essential services. The existing diesel generators remain available and unaltered. AMBESS is intended to minimise changes to the vessel's existing essential-power architecture; the final electrical integration, protection philosophy and applicable class approval requirements are established for each vessel and installation.

The system is sized to the vessel, in modules, against the measured load — not supplied as one arrangement that suits every hull.

And the question every port asks first

Fire.

Every cell is immersed in dielectric fluid, providing direct thermal contact around the cell surfaces and a continuous path for heat removal. The architecture is intended to limit temperature gradients and reduce the conditions that can contribute to cell-to-cell thermal propagation. Its performance against thermal propagation will ultimately be established through the applicable module and system testing and the marine type-approval programme.

A vessel cannot be evacuated and there is no fire brigade alongside, so we treat containment as a problem of architecture rather than one to be solved by adding suppression equipment.

The fluid is a synthetic ester — a K-class less-flammable insulating liquid with a fire point of 306 °C against roughly 140–160 °C for a mineral oil, **92.91 per cent readily biodegradable at 28 days** under OECD 301B, with acute aquatic toxicity data to OECD 201, 202 and 203, and IEC 61099 conformity. A leak into river or coastal water is not the pollution event a hydrocarbon would be.

On certification, plainly: the battery module is presently undergoing marine classification type approval. It is not yet certified and we do not describe it as such. The cell holds a full IEC 62619:2022 test report under the IECCE CB Scheme, together with UL 1642, UL 1973 and a UL 9540A cell-level report — those are the cell manufacturer's certificates, for the cell.

What this is worth on your own vessel

We do not know, and neither does anyone who has not measured it.

The number can only be established by measurement, because anchor and low-load generator hours are rarely isolated from the vessel's overall operating fuel figure.

Give us one vessel and its fuel figures. We will survey it, measure the hotel load and the duty cycle at our own cost, and put the arithmetic on a single page — the hours, the litres, the money, and what it would take. If the answer is that it is not worth doing on that hull, we will say so.

We would rather be told we are wrong about one boat than be believed about a fleet.

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Sources. Time at anchorage and the auxiliary share of fuel consumed there: International Maritime Organization, *Just In Time Arrival Guide* (global AIS analysis), and Kanellos et al., *Defining the Power and Energy Demands from Ships at Anchorage for Offshore Power Supply Solutions*, *Energies* 18(7):1766, 2025 (EU BlueBARGE project). · Engine efficiency, low-load operation and auxiliary load management: IMO, *Energy Management Onboard*, Module 4, sections 6.2.1 and 6.2.3. · Auxiliary share of total fuel consumption: IMO GreenVoyage2050, Energy Efficiency Technologies Information Portal. · Deep-sea trawler consumption: Indian Council of Agricultural Research, economic analysis of 22 m and 23 m deep-sea trawlers. · Research vessel endurance planning: UNOLS. · Fluid and cell evidence: ERDA RP-2324-036777 (IEC 61099); Ignito I231211009-1 (EN 45545-2, Class K, fire point 306 °C); Intertek (OECD 301B); OECD 201, 202 and 203 reports; TUV Rheinland CN2437VS 001 (IEC 62619:2022, IECCE CB Scheme); UL 1642 MH64633; UL 1973 UL-US-2336292-0; UL 9540A cell-level report. · The 169-day operating figures are our own, from the platform's written register. Copies of any item are available on request.